

Parameter Estimation

Preliminary minimax analysis via reduction to *Gaussian location mixtures*.

$$\mathcal{A}_k := \left\{ \frac{1}{k} \sum_{i=1}^k \delta_{b_i} : b_1, \dots, b_k \in \mathbb{R} \right\} \quad \bar{\mathcal{G}}_k := \{s * \rho : s \in \mathcal{A}_k\}$$
$$\rho = \mathcal{N}(0,1)$$

Parameter Estimation

Theorem (M, Manole, Rigollet, forthcoming)

Observe $Y_i = f_\mu(X_i) + \epsilon_i$, $i = 1, \dots, n$ with $\epsilon_i \sim \mathcal{N}(0,1)$, $X_i \sim \mathcal{U}([0,1])$ independently. Set $Q_{f_\mu} = \text{Law}(X_1, Y_1)$. We define the minimax risk,

$$\mathcal{R}_n = \inf_{\hat{\mu}_n} \sup_{\mu \in \bar{G}_k} \mathbb{E} [W_1(\hat{\mu}_n, \mu)]$$

in expectation over $Q_{f_\mu}^{\otimes n}$. There exists $C_k > 0$ such that,

$$\frac{1}{C_k} \left(\frac{1}{n} \right)^{\frac{1}{2k}} \leq \mathcal{R}_n \leq C_k \left(\frac{\log^2(n)}{n} \right)^{\frac{1}{2k}}$$

Upper bound: achieved by square loss ERM for $f_\mu(\cdot)$.